

Semiconductor Characteristics Course

Page 1 — Questions 1–14

1) What do you understand by electron energy loss spectroscopy? Explain working principle of electron energy loss spectroscopy.

Definition (what it is):

EELS is a method to find what atoms and bonds are inside a **very thin** sample. We send fast electrons through the sample and **measure how much energy they lose**. The amount of energy lost acts like a **fingerprint** of the material.

Working principle (step by step):

1. **Make a very thin sample** (like a tiny slice) so electrons can pass through it.
2. **Send a narrow beam of fast electrons** through the slice (this is usually done inside a TEM/STEM).
3. **Inside the sample, some electrons lose a little energy** when they interact with atoms (they can shake the atoms' electrons or knock an inner electron to a higher level).
4. **After the sample, an energy filter/spectrometer separates the electrons** by how much energy they lost.
5. **A detector counts them** and draws a graph: intensity vs energy loss (the **EELS spectrum**).
6. **Read the spectrum:**
 - Peaks at certain energy losses tell **which elements** are present.
 - Small shifts/shapes of peaks tell **how those atoms are bonded** (for example, metal vs oxide).
 - The low-loss part can also tell **thickness** and sometimes **bandgap**.
 - When we scan the beam point-by-point, we can make **element maps** at very small (nanometer/atomic) scales.

2) Define SEM. Explain working principle of SEM. What are the applications and limitations of SEM? What are the differences between SEM and STM structural characteristics?

☐ **Definition (what is SEM)** Scanning Electron Microscopy

SEM is a microscope that uses **electrons instead of light** to look at the **surface** of a material.

It shows **very detailed pictures** of the surface — like hills, cracks, or particles — that normal microscopes cannot see.

☐ **Working principle (step by step)**

1. **Electron gun makes electrons** – The microscope has a source that produces a stream of fast electrons.
 2. **Electrons are focused** – Special lenses (magnetic) make the electron beam very narrow and sharp.
 3. **Beam scans the surface** – The beam moves line by line over the sample (like how a printer moves).
 4. **Electrons hit the surface** – When the beam touches the surface, the atoms release other electrons and signals.
 5. **Detectors collect signals** – The released (secondary or backscattered) electrons are collected by detectors.
 6. **Computer makes an image** – The signals are turned into a clear image that shows the shape and texture of the surface.
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☐ **Applications (where we use SEM)**

- To see **surface shapes** of metals, powders, thin films, and biological samples.
 - To study **cracks, grains, and coating quality**.
 - To do **failure analysis** — find why a material broke or failed.
 - To check **particle size** and **surface contamination**.
 - With an **EDX detector**, we can also find **what elements** are on the surface.
Energy Dispersive X-ray Spectroscopy
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☐ **Limitations (what SEM cannot do easily)**

- Works only in a **vacuum**, so **wet or living samples** are hard to study.

Scanning Electron Microscope

- **Non-conducting materials** (like plastic or glass) must be coated with a thin metal layer, otherwise the image gets blurry.
- It mostly shows the **surface**, not the inside of the sample.
- **High electron energy** can damage soft or thin samples.

Feature	SEM	STM Scanning Tunneling Microscope
Type of image	Shows surface shape and texture	Shows atoms and electronic structure
Working principle	Uses electron beam hitting the surface	Uses tunneling current between a sharp tip and the surface
Sample type	Can study metals, ceramics, and coated insulators	Needs conductive or semiconductive samples only
Resolution	Up to a few nanometers	Can reach atomic resolution
Image type	3D-like image of surface	Atomic-scale map of electron density

3) Why does STM require conducting or semiconducting samples? What is the typical resolution achievable with STM?

Scanning Tunneling Microscope (STM) — very simple explanation

☐ Why STM needs conducting or semiconducting samples

1. STM works by passing a **tiny electric current** between a very **sharp metal tip** and the **sample surface**.
 2. The current flows through **quantum tunneling**, which only happens if **electrons can move** between the tip and the sample.
 3. **Conductors** (like metals) and **semiconductors** allow electrons to move easily — so tunneling current can flow.
 4. **Insulators** do not let electrons move freely — so **no current** flows and STM **cannot make an image**.
 5. That's why STM needs a **conductive or semiconductive surface** for the tip to “feel” it.
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□ How STM makes an image (simple steps)

1. The tip comes very close to the surface (just a few atoms away).
 2. A **tiny voltage** is applied between the tip and surface.
 3. **Electrons tunnel** across the small gap → very small current flows.
 4. The **computer adjusts the tip height** to keep the current constant as the tip moves over the surface.
 5. The up-and-down motion of the tip forms a **map of the atoms** on the surface.
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□ Typical resolution of STM

- **Side-to-side (lateral) resolution:** about **0.1 nanometers (nm)** — this means STM can see **individual atoms**.
- **Up-and-down (height) resolution:** about **0.01 nanometers (nm)** — it can detect very tiny changes in height.

□ In short:

STM needs conducting or semiconducting materials **because electrons must flow** between the tip and the sample. It can see **atoms one by one**, with **very high resolution** — one of the best among all microscopes.

4) What do you understand by DLTS? What information can be obtained about defect density from DLTS spectra? How is DLTS used to study defects in perovskite and silicon solar cells?

Deep Level Transient Spectroscopy (DLTS) — very simple explanation

□ Definition (what is DLTS)

DLTS is a **measurement technique** used to find **defects or traps** inside **semiconductors**. These defects are like “tiny holes” in the crystal that can capture or release charge carriers (electrons or holes). DLTS helps us know **how many defects** are there and **how deep** they are in energy.

□ Working principle (step by step)

1. A semiconductor device (like a **diode or solar cell**) is placed under test.
 2. A **voltage pulse** is applied to fill the traps with charge.
 3. After the pulse, the voltage is removed, and the **trapped charges start to leave** the traps with time.
 4. This release causes a small **change in capacitance** of the device.
 5. The system records this **capacitance change (transient)** as temperature slowly increases.
 6. A **DLTS spectrum** is made from this data — it shows peaks at temperatures where certain traps release their charges.
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□ Information from DLTS spectra

- **Defect density:** the **height or area of a DLTS peak** shows how many defects (traps) are in the sample.
 - **Activation energy:** the **position** of the peak (temperature) tells how deep the trap lies in the energy gap.
 - **Capture cross-section:** tells how easily the defect captures charge carriers.
 - **Type of trap:** by analyzing the sign and shape of peaks, we know if the trap catches **electrons or holes**.
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☀□ Use of DLTS in perovskite and silicon solar cells

- DLTS helps find **defects that lower efficiency**, such as traps causing recombination of charge carriers.
 - In **silicon solar cells**, DLTS shows **impurities and defect levels** that reduce lifetime or open-circuit voltage.
 - In **perovskite solar cells**, DLTS helps locate **trap positions** at interfaces or grain boundaries that cause **instability or degradation**.
 - It is also used to check **how defects change** after **light, heat, or moisture exposure**.
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□ In short:

DLTS is a simple and powerful way to **see hidden defects** inside semiconductors. It tells us **how many, how deep, and what kind** of traps are present — very useful for improving **solar cell performance and stability**.

5) Define C-V characterization. Explain types of C-V measurement technique. How does DLTS differ from conventional capacitance–voltage (C–V) profiling?

Capacitance–Voltage (C–V) Characterization — very simple explanation

□ Definition (what is C–V characterization)

C–V characterization is a **test method** used to study **semiconductor devices** like **diodes** and **MOS capacitors**. It measures **how the capacitance changes** when we apply **different voltages**. From this, we can learn about **doping levels**, **interface charges**, and **electric field behavior** inside the device.

□ Working principle (step by step)

1. A small **AC signal** (tiny changing voltage) is applied on top of a **DC bias voltage**.
 2. As the DC voltage changes, the **depletion region** in the semiconductor becomes wider or thinner.
 3. When the depletion width changes, the **capacitance also changes**.
 4. The C–V meter measures this change and draws a **graph of capacitance vs voltage**.
 5. The shape of this graph gives important information about the **material and device quality**.
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□ Types of C–V measurement techniques

1. **High-frequency C–V:**
 - Uses signals around 1 MHz.
 - Measures capacitance without letting slow traps or interface states respond.
 - Common for MOS and diode studies.
 2. **Low-frequency or quasi-static C–V:**
 - Uses a slower signal so even traps can follow the change.
 - Shows information about **interface states** and **slow defects**.
 3. **C–V profiling:**
 - Used to find **doping concentration vs depth**.
 - From the slope of a **$1/C^2$ vs voltage** graph, we calculate the **doping profile** inside the semiconductor.
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□ Difference between DLTS and C–V profiling

Feature	C–V	DLTS
What it measures	Capacitance vs voltage (steady-state)	Capacitance vs time after voltage pulse (transient)
Information obtained	Doping, interface charge, built-in voltage	Deep traps/defects, their energy and density
Speed of measurement	Slow, static	Fast, time-based
Purpose	Surface and doping analysis	Defect and trap study

□ In short:

C–V characterization tells how **charges and doping** are arranged in a semiconductor. DLTS, on the other hand, looks at **how defects trap and release charges** with time and temperature.

Both use capacitance, but they give **different kinds of information** about the material's quality.

6) What is Atomic Force Microscopy (AFM), and how does it differ from Scanning Electron Microscopy (SEM)? Explain working principle of atomic force microscopy. Mention some applications of AFM.

Atomic Force Microscopy (AFM) — very simple explanation

□ Definition (what is AFM)

AFM is a **type of microscope** that can **feel the surface** of a material instead of looking at it with light or electrons. It uses a **very sharp tip** to touch or move just above the surface and measures **tiny forces** between the tip and the surface. From these forces, it makes a **3D image** of the surface at the **nanometer or atomic level**.

□ Working principle (step by step)

1. The microscope has a **sharp tip** fixed at the end of a small **spring-like arm** called a **cantilever**.
 2. The tip moves **over the surface** of the sample.
 3. When the tip gets very close, **tiny forces** (like attraction or repulsion) make the cantilever bend slightly.
 4. A **laser beam** shines on the back of the cantilever and reflects to a **photodiode detector**.
 5. As the cantilever bends, the reflected laser moves — this tells how much the tip is bending.
 6. A **computer records the tip movement** and creates a **3D map** showing the surface height and shape.
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□ Main modes of AFM operation

- **Contact mode:** tip touches the surface lightly while scanning.
 - **Tapping mode:** tip touches the surface only sometimes — good for soft samples.
 - **Non-contact mode:** tip stays just above the surface and measures attractive forces.
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□ Difference between AFM and SEM

Feature	AFM	SEM
Type of signal	Measures forces with a mechanical tip	Uses electrons to scan the surface
Image type	Gives 3D height image	Gives 2D image with shading
Environment	Can work in air, liquid, or vacuum	Needs vacuum
Sample type	Works for insulators and conductors	Non-conductors often need metal coating
Damage	Gentle, little damage	Can damage soft materials

□ Applications of AFM

- Measuring **surface roughness** of thin films.
 - Studying **nanoparticles, fibers, and coatings**.
 - Checking **defects, cracks, or grains** in materials.
 - Measuring **mechanical properties** like stiffness or adhesion.
 - Imaging **biological samples** such as cells or DNA in air or liquid.
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□ In short:

AFM “feels” the surface using a tiny tip to make a **3D image**. It is very useful for both **hard and soft materials**, works without vacuum, and shows **surface features at the nanometer scale** — something SEM cannot always do.

7) How can AFM be used to measure surface roughness of thin films? How do environmental conditions influence AFM measurements?

AFM for Measuring Surface Roughness — very simple explanation

□ How AFM measures surface roughness of thin films

1. **AFM scans the surface** using a very sharp tip that moves across the thin film.
 2. The tip **moves up and down** as it passes over bumps and valleys on the surface.
 3. The microscope **records these height changes** and creates a **3D image** of the surface.
 4. From this image, software calculates **roughness values**, such as:
 - **Ra (Average Roughness)**: the average height of bumps and dips.
 - **Rq (Root Mean Square Roughness)**: shows how rough the surface is overall.
 5. By scanning different areas and comparing Ra or Rq, we can tell if a film is **smooth or uneven**.
 6. These roughness values are important because surface smoothness affects **optical, electrical, and coating** properties of thin films.
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□ How environmental conditions affect AFM measurements

1. **Vibration:**

- Any small vibration from the floor or nearby machines can **shake the tip**, making the image blurry.
- AFM is usually placed on a **vibration-free table**.

2. **Temperature:**

- Changes in temperature cause the sample or AFM parts to **expand or contract**, shifting the image slowly.
- It is best to keep the temperature **stable**.

3. **Humidity (moisture):**

- Water in the air can form a **thin layer** between the tip and surface.
- This layer can **pull the tip** and affect measurements — especially in contact mode.
- Using **tapping mode** or **dry air** helps reduce this problem.

4. **Dust or contamination:**

- Dust particles on the surface can **stick to the tip** or give false bumps.
- Samples and tips must be **kept clean**.

□ In short:

AFM measures surface roughness by **scanning the surface** and recording tiny height changes.

For accurate results, the environment must be **clean, vibration-free, and stable**, so the tip can move smoothly and measure the true roughness of the thin film.

8) What is Transmission Electron Microscopy (TEM), and how does it differ from Scanning Electron Microscopy (SEM)? Why must TEM samples be extremely thin? How can TEM be used to determine the crystal structure of materials?

Transmission Electron Microscopy (TEM) — very simple explanation

□ Definition (what is TEM)

TEM is a **microscope** that uses a beam of **fast electrons** to look **through a very thin sample**.

It shows the **inside structure** of materials — such as **grains, layers, and even atoms**. It gives much higher resolution than an optical or scanning electron microscope.

□ Working principle (step by step)

1. A powerful **electron gun** sends high-energy electrons toward the sample.
 2. The electrons **pass through** the thin sample (not just bounce off the surface).
 3. Some electrons are **scattered** or **transmitted** depending on the internal structure.
 4. **Lenses focus** the transmitted electrons to form an image on a screen or detector.
 5. The image shows **internal details** like crystal patterns, defects, or layers.
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□ Difference between TEM and SEM

Feature	TEM	SEM
Type of image	Shows internal structure (inside the sample)	Shows surface structure
How electrons work	Electrons pass through the sample	Electrons bounce off the surface
Sample requirement	Must be extremely thin (less than 100 nm)	Can be thick or bulk sample
Resolution	Very high (can see atoms)	Lower (few nanometers)
Image type	Flat 2D projection of internal parts	3D-like surface image

□ Why TEM samples must be extremely thin

- Electrons must **pass through the sample** to make an image.
 - If the sample is thick, electrons will be **scattered or absorbed**, and no clear image will form.
 - Thin samples (usually **less than 100 nanometers**) allow electrons to transmit properly, giving a **sharp and clear image**.
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□ How TEM is used to find crystal structure

1. TEM can do **Selected Area Electron Diffraction (SAED)**:
 - The electron beam passes through a small area.
 - The electrons form **diffraction spots or rings** on a screen.
 - The pattern tells about the **lattice spacing and symmetry** of the crystal.
 2. **High-Resolution TEM (HRTEM)**:
 - Directly shows **atomic planes** as bright and dark lines.
 - Measuring the distance between these lines gives **crystal plane spacing**.
 - Helps identify **crystal orientation and defects**.
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□ In short:

TEM lets us see **inside a material at the atomic level** by sending electrons **through a very thin sample**. It is different from SEM, which only shows the **surface**.

TEM helps scientists understand **crystal structure, grain boundaries, and defects** in materials very clearly.

9) What is Energy-Dispersive X-ray Spectroscopy (EDX), and what is its working principle? How is EDX applied in the analysis of thin films and nanomaterials?

Energy-Dispersive X-ray Spectroscopy (EDX or EDS) — very simple explanation

□ Definition (what is EDX)

EDX is a **technique used with SEM or TEM** to find out **which elements** are present in a material.

It works by measuring the **X-rays** that come out of the sample when it is hit by an **electron beam**.

Each element gives off X-rays with its **own special energy**, like a fingerprint.

□ Working principle (step by step)

1. A focused **electron beam** hits the surface (in SEM) or passes through (in TEM) the sample.
 2. The incoming electrons **knock out inner-shell electrons** from the atoms in the sample.
 3. When an outer electron falls into that empty place, it releases an **X-ray**.
 4. The **energy of this X-ray** depends on the element — each element gives a different energy value.
 5. A **detector** measures these X-rays and makes a **spectrum (graph)** of X-ray intensity versus energy.
 6. The **peaks in the graph** tell us which elements are present, and the **height of peaks** shows their relative amounts.
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□ Application of EDX in thin films and nanomaterials

1. **Element identification:**
 - EDX can find which **elements** are present in a thin film or nanoparticle.
 - For example, it can check if a perovskite film contains **lead, iodine, or bromine** correctly.
2. **Composition check:**
 - It helps measure the **ratio of elements** (like metal-to-oxygen ratio) to confirm proper film composition.
3. **Element mapping:**

- By scanning point-by-point, EDX can make a **color map** showing **where each element** is located in the sample.
- Useful for checking **uniformity** of thin films or **core-shell** structure in nanoparticles.

4. **Detecting impurities:**

- EDX can find **unwanted elements** or **contaminants** that might affect material performance.

☐ **In short:**

EDX is a simple and powerful tool that tells **what elements** are inside a material. It works by measuring the **energy of X-rays** released after the electron beam hits the atoms. In thin films and nanomaterials, EDX helps check **composition, purity, and element distribution**, which are very important for material quality.

10) What is Auger Electron Spectroscopy (AES)? How can AES be used to determine the elemental composition of a surface? What information can AES provide about chemical bonding states?

Auger Electron Spectroscopy (AES) — very simple explanation

☐ **Definition (what is AES)**

AES is a **surface analysis technique** that tells us **which elements** are present on the **top few nanometers** of a material. It works by studying the **energy of electrons** (called *Auger electrons*) that come out from the surface when it is hit by an **electron beam**. Because the Auger electrons come only from the very top layer, AES is **extremely surface-sensitive**.

☐ **Working principle (step by step)**

1. A focused **electron beam** hits the surface of the sample.
2. The beam **knocks out an inner-shell electron** from an atom, leaving an empty space.
3. Another electron from a higher level fills this empty space.

4. During this process, a third electron is released from the atom — this is the **Auger electron**.
 5. The **energy of this Auger electron** depends on the type of element that emitted it.
 6. A **detector measures** the energies of all the emitted electrons and makes a **spectrum** (energy vs intensity).
 7. Each element has **unique energy peaks**, so by reading the peaks, we can tell **which elements** are present.
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□ **How AES determines elemental composition**

- The **energy position** of each peak identifies the **element**.
 - The **height or area** of each peak shows **how much** of that element is on the surface.
 - Because the electrons come only from the **top 1–3 nanometers**, AES gives a **true surface composition**.
 - By removing layers slowly using **ion sputtering**, AES can also show **composition with depth** (depth profiling).
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□ **What AES tells about chemical bonding states**

- The **exact energy** of the Auger peaks can **shift slightly** depending on how the atom is bonded.
 - For example, the same element (like copper) will show **different peak positions** in **metallic Cu** and in **CuO (oxide)**.
 - This helps identify **oxidation states**, **chemical bonds**, and **surface reactions**.
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□ **In short:**

AES uses an **electron beam** to make and measure **Auger electrons**, which come from the top surface of a material. It helps find **which elements** are on the surface and gives extra details about their **chemical bonding** and **oxidation states** — making it a powerful tool for studying surfaces and thin films.

11) How can AES help in understanding degradation mechanisms in perovskite solar cells? What are the main challenges of using AES for insulating materials?

Auger Electron Spectroscopy (AES) in Perovskite Solar Cells — very simple explanation

□ How AES helps to understand degradation in perovskite solar cells

1. **Perovskite solar cells** can lose performance over time due to **moisture, heat, light, or oxygen**.
 2. AES studies the **surface composition** and tells **which elements have changed** after ageing.
 3. It can detect **loss of halides** (like iodine or bromine) and **formation of new compounds** such as **PbI₂**.
 4. AES can **map elements** at different spots (for example, at **grain boundaries or interfaces**) to find where changes or reactions happen first.
 5. By comparing a **fresh cell** and an **aged cell**, AES helps identify which **chemical changes** cause **degradation** or **instability**.
 6. This information helps scientists improve **material stability** and **protective layers** in perovskite solar cells.
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□ Main challenges of using AES for insulating materials

1. **Charging problem:**
 - Insulators do not conduct electricity.
 - When the electron beam hits the surface, **charges build up**, which **shifts or distorts** the AES signals.
2. **Signal distortion:**
 - The **charging** can make peaks move or become unclear, giving **wrong results**.
3. **Sample damage:**
 - Continuous electron beam exposure can **damage** sensitive materials like perovskites.
4. **Solutions:**
 - Use **low beam energy** or **lower current**.
 - Apply a **very thin conductive coating**.
 - Use **charge neutralizers** (like electron or ion flood guns) to remove excess charge.

□ **In short:**

AES helps find **what chemical changes** occur during the **degradation of perovskite solar cells** by studying the **surface elements**. However, since perovskite is partly **insulating**, AES faces **charging and damage problems**, which must be controlled carefully to get correct results.

12) Define photoluminescence (PL) and describe its fundamental mechanism. How does photoluminescence spectroscopy help in determining the bandgap of semiconductors?

Photoluminescence (PL) — very simple explanation

□ **Definition (what is PL)**

Photoluminescence (PL) is a process in which a material **absorbs light** and then **gives off light**. It is a **non-destructive** way to study the **electronic and optical properties** of semiconductors.

□ **Fundamental mechanism (step by step)**

1. A **light source** (usually a laser) shines on the semiconductor.
 2. The incoming photons give energy to **electrons**, moving them from the **valence band** to the **conduction band**.
 3. This creates **electron-hole pairs** — the electron is free, and the hole is left behind.
 4. After a short time, the **electron falls back** to the valence band and **recombines** with the hole.
 5. When this happens, the material **emits light** (a photon).
 6. The **energy (or color)** of this emitted light depends on the **energy difference** between the two bands.
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□ How PL helps to find the bandgap of semiconductors

1. The **energy of the emitted light** is almost equal to the **bandgap energy** of the material.
2. By measuring the **wavelength** of the strongest PL peak, we can calculate the **bandgap** using:

$$E(eV) = \frac{1240}{\text{wavelength (nm)}}$$

3. For example, if the PL peak is at 620 nm,

($E = 1240 / 620 = 2.0 \text{ eV}$) → that is the bandgap energy.

4. The **shape** of the PL spectrum can also show if there are **defects** or **impurities**, because they create extra peaks at lower energies.
5. In this way, PL gives information about **bandgap**, **defect levels**, and **material quality**.

□ In short:

Photoluminescence means a material **shines light after absorbing light**. It happens when excited electrons **fall back** to lower energy levels. By measuring the **energy or color** of that emitted light, we can easily **find the bandgap** and **check the purity** of semiconductors.

13) Discuss the applications of photoluminescence in optoelectronic device characterization. What is Fourier Transform Infrared (FTIR) spectroscopy, and how does it work?

Photoluminescence (PL) and Fourier Transform Infrared (FTIR) Spectroscopy — very simple explanation

□ Applications of Photoluminescence (PL) in optoelectronic devices

1. Bandgap measurement:

- PL helps find the **bandgap energy** of materials used in LEDs, lasers, and solar cells.
- The main PL peak position shows the **energy difference** between conduction and valence bands.

2. Defect detection:

- Extra PL peaks at lower energy show the presence of **defects or impurities**.
- Helps check **material quality** and **purity**.

3. Carrier lifetime study:

- Time-resolved PL measures **how long electrons stay excited** before recombining.
- Longer lifetime means **better quality** and **less recombination loss**.

4. Uniformity check:

- PL mapping can show if the material surface is **uniform or uneven** in composition.
- Useful for large wafers or thin films.

5. Interface and surface studies:

- PL can show **energy loss** at interfaces, helping improve **junction quality** in solar cells and LEDs.
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□ Fourier Transform Infrared (FTIR) Spectroscopy — what it is

Definition:

FTIR is a technique used to find the **chemical bonds and functional groups** in a material by studying how it **absorbs infrared (IR) light**.

□ Working principle (step by step)

1. A beam of **infrared light** passes through or reflects from the sample.
2. The sample **absorbs certain IR wavelengths** that match the vibration of its chemical bonds.
3. The remaining light goes into a **Michelson interferometer**, which creates a mixed light signal called an **interferogram**.
4. A **detector records** this interferogram.
5. A computer then uses a **Fourier transform** to separate and display the data as a **spectrum** (absorbance vs wavelength).
6. Each **peak** in the spectrum shows a specific **bond vibration** — for example:
 - O–H bond → around 3400 cm^{-1}
 - C=O bond → around 1700 cm^{-1}

□ In short:

- **PL** helps understand **optical and electronic quality** of materials by showing how they emit light.
- **FTIR** helps identify **chemical bonds and groups** by showing how they absorb infrared light.
- Both are important tools for studying and improving **optoelectronic materials** like semiconductors and thin films.

14) How does FTIR differ from conventional dispersive infrared spectroscopy? What information can be obtained from an FTIR spectrum?

Fourier Transform Infrared (FTIR) Spectroscopy — very simple explanation

□ Difference between FTIR and conventional dispersive infrared spectroscopy

Feature	FTIR (Fourier Transform Infrared)	Dispersive Infrared Spectroscopy
How it works	Measures all infrared wavelengths at once and then uses a computer to separate them with a Fourier transform	Measures one wavelength at a time using a monochromator
Speed	Very fast (takes only a few seconds)	Slow , because it scans each wavelength separately
Accuracy	Very high wavelength accuracy (uses a laser reference)	Less accurate due to mechanical scanning errors
Signal strength	Collects more light at once , so it has better signal-to-noise ratio	Collects less light, so signal is weaker
Maintenance	Fewer moving parts → more stable	More moving parts → more alignment needed

In short, FTIR is **faster, more accurate, and more sensitive** than the old dispersive method.

□ Information obtained from an FTIR spectrum

1. Chemical bonds and functional groups:

- Each peak in the spectrum corresponds to a specific **bond vibration**.
- Example:
 - O–H bond → around 3400 cm^{-1}
 - C=O bond → around 1700 cm^{-1}
 - C–H bond → around 2900 cm^{-1}

2. Material identification:

- The pattern of peaks acts like a **fingerprint** to identify unknown materials.

3. **Chemical composition and impurities:**

- Extra or missing peaks show **impurities** or **changes in composition**.

4. **Bonding and structure changes:**

- Peak shifts or intensity changes can indicate **chemical reactions, oxidation, or phase changes**.

5. **Film thickness or concentration (in thin films or solutions):**

- Peak **area or height** can be used to measure **amount** of a substance.

□ **In short:**

FTIR is a **modern, faster, and more accurate** version of infrared spectroscopy. From its spectrum, we can learn about **chemical bonds, functional groups, purity, and structure** of a material — making it very useful in **chemistry, materials science, and semiconductor research**.

Page 2 — Assignment (Questions 1–10)

1) What do you understand by optical transmittance and reflectance spectroscopy? How can you calculate optical transmittance and reflectance spectroscopy for material?

Optical Transmittance and Reflectance Spectroscopy — very simple explanation

☒ Definition (what it means)

Optical transmittance and reflectance spectroscopy are **simple light-based methods** used to study how a material **interacts with light**.

- **Transmittance** means how much light **passes through** a material.
- **Reflectance** means how much light **bounces back** from the surface.

By measuring both, we can understand the material's **optical properties**, like transparency, absorption, and surface quality.

☐ Working principle (step by step)

1. A **light source** (like a lamp or laser) shines on the material.
 2. Some light is **transmitted** through the material.
 3. Some light is **reflected** back from the surface.
 4. The rest of the light is **absorbed** inside the material.
 5. Detectors measure how much light comes out as transmitted and reflected.
 6. A **computer plots a graph** (intensity vs wavelength) — this is the optical spectrum.
-

🔍 How to calculate transmittance and reflectance

📊 How to calculate transmittance and reflectance

- Transmittance (T):

$$T = \frac{I_t}{I_0} \times 100\%$$

where:

I_t = transmitted light intensity

I_0 = incident (incoming) light intensity

- Reflectance (R):

$$R = \frac{I_r}{I_0} \times 100\%$$

where:

I_r = reflected light intensity

I_0 = incident light intensity

- Sometimes, we also find **Absorbance (A)** using:

$$A = 1 - (R + T)$$

This shows how much light is **absorbed** by the material.

🔍 Uses of optical transmittance and reflectance spectroscopy

- To check **optical quality** and **transparency** of thin films, glasses, and coatings.
 - To find **bandgap energy** by analyzing how light is absorbed.
 - To study **surface reflection** and **color** properties of materials.
 - To monitor **film thickness** and **uniformity**.
-

🔍 In short:

Optical transmittance and reflectance spectroscopy tell us **how much light passes through or reflects** from a material. By measuring light intensity before and after hitting the material, we can easily calculate **T** and **R**, which help us understand the material's **optical behavior and quality**.

2) What do you understand by ellipsometry? Explain basic principle of ellipsometry.

Ellipsometry — very simple explanation

📖 Definition (what is ellipsometry)

Ellipsometry is a **non-destructive optical technique** used to measure the **thickness** and **optical properties** (like refractive index and absorption) of **thin films**.

It works by studying how **polarized light** changes after it **reflects** from the surface of a material.

□ Basic principle (step by step)

1. A **beam of light** with known polarization is directed onto the surface of a sample at a specific angle.
 - *Polarized light* means the light waves vibrate in a specific direction.
 2. When this light **reflects** from the surface, its **polarization and intensity change** depending on the film's properties.
 - If the film is thick or thin, or has different refractive index, the reflected light changes differently.
 3. The reflected light is then passed through an **analyzer**, which measures how the **polarization has changed**.
 4. The instrument records two main values:
 - **Ψ (Psi):** change in amplitude (strength of light).
 - **Δ (Delta):** change in phase (timing of light waves).
 5. By comparing the measured Ψ and Δ with computer models, we can calculate:
 - **Film thickness** (how thick the layer is).
 - **Refractive index (n)** and **extinction coefficient (k)** (how light moves and absorbs in the film).
-

📖 Why ellipsometry is useful

- It can measure **very thin films** — even just a **few nanometers** thick.
 - It does **not damage** the sample.
 - It gives **highly accurate** data about **optical and surface properties**.
-

📖 In short:

Ellipsometry works by shining **polarized light** on a surface and checking **how that light changes after reflection**.

From those changes, we can find the **thickness** and **optical constants** of thin films — making it a powerful tool for studying coatings, semiconductors, and nanomaterials.

3) What is Auger Electron Spectroscopy (AES)? How can AES be used to determine the elemental composition of a surface?

Auger Electron Spectroscopy (AES) — very simple explanation

📖 Definition (what is AES)

Auger Electron Spectroscopy (AES) is a **surface analysis method** that helps to find out **which elements** are present on the **very top layer** of a material (only a few nanometers deep). It works by using an **electron beam** to make atoms in the surface give off special electrons called **Auger electrons**. By studying the energy of these electrons, we can tell what elements are on the surface.

□ Working principle (step by step)

1. A focused **electron beam** hits the surface of the sample.
 2. This beam **knocks out an inner electron** from one of the atoms, leaving a small empty spot (a “hole”).
 3. An electron from a higher energy level drops down to fill the hole.
 4. While this happens, a second electron is released from the atom — this is the **Auger electron**.
 5. The **energy of this Auger electron** depends on the type of element it came from.
 6. A **detector** measures the energy of all emitted electrons and creates a **spectrum** (a graph of energy vs number of electrons).
 7. Each element has **unique energy peaks**, so the peaks tell us which elements are present on the surface.
-

📖 How AES is used to find surface composition

- Each **peak position** in the AES spectrum identifies a **specific element**.
 - The **height or area** of the peak shows **how much** of that element is on the surface.
 - Because Auger electrons come from only the **top few nanometers**, AES gives **true surface information**.
 - By slowly removing thin layers (using ion sputtering), AES can also show how the **element concentration changes with depth** — this is called **depth profiling**.
-

📌 In short:

AES uses an **electron beam** to make atoms on the surface release **Auger electrons**. By measuring their energy, we can find out **which elements** are on the surface and their **amount**, making AES a powerful tool for **surface and thin film analysis**.

4) What information can AES provide about chemical bonding states?

Chemical Bonding Information from Auger Electron Spectroscopy (AES) — very simple explanation

📌 Basic idea

AES not only tells **which elements** are on a surface, but it can also give information about **how those elements are bonded** — this is called the **chemical bonding state** or **chemical environment**.

In simple words, AES can show whether an element is in **metal form**, **oxide form**, or **combined with other atoms**.

❑ How AES gives bonding information (step by step)

1. When AES measures the energy of Auger electrons, the **exact energy** depends slightly on the **chemical environment** of the atom.
 2. If the atom is bonded to **different elements** (like oxygen, sulfur, or chlorine), the **electron energy levels shift** a little.
 3. These small **energy shifts** or **changes in peak shape** are seen clearly in the AES spectrum.
 4. By comparing the measured spectrum to known reference spectra, we can identify **how the atom is bonded**.
-

📌 Examples of what AES can show

- **Metal vs oxide:**
 - Pure copper (Cu metal) and copper oxide (CuO) have **slightly different Auger peak energies**.
 - AES can tell which one is present on the surface.
- **Chemical reactions:**
 - If a surface reacts with air or another element, AES can detect new **chemical bonds** forming.
- **Surface contamination:**
 - AES can reveal if carbon, oxygen, or other atoms are **chemically attached** to the surface material.

📌 In short:

AES detects **tiny energy changes** in Auger peaks that happen when atoms form **different chemical bonds**.

By studying these shifts, AES can tell the **chemical state** of elements — for example, whether an element is **metallic, oxidized, or part of a compound** — giving a clear picture of the surface chemistry.

5) How can AES help in understanding degradation mechanisms in perovskite solar cells?

Use of Auger Electron Spectroscopy (AES) in Understanding Degradation of Perovskite Solar Cells — very simple explanation

📌 Basic idea

Perovskite solar cells can lose their efficiency over time because of **moisture, heat, oxygen, or light**.

AES helps scientists **study the surface chemistry** and **find out what changes** happen during this degradation process.

📌 How AES helps (step by step)

1. Before and after comparison:

- AES can analyze a **fresh solar cell** and an **aged or damaged one**.
- By comparing both, we can see which elements have **changed or disappeared** on the surface.

2. Element identification:

- AES tells which **elements** (like lead, iodine, bromine, carbon, oxygen) are present on the surface.
- If some elements are missing or new ones appear, it means a **chemical reaction** has occurred.

3. Detection of degradation products:

- AES can detect the formation of **PbI₂** or other unwanted compounds that appear when perovskite **breaks down**.
- It can also show **halide loss** (loss of iodine or bromine), which is a key reason for cell failure.

4. Element migration:

- AES can **map elements** across the surface or layer interface.
- It shows if ions (like iodine or lead) **move or gather** at grain boundaries or contacts, which causes instability.

5. Environmental effects:

- AES can test samples exposed to **light, heat, or humidity** to see how these conditions change surface chemistry.
-

📌 In short:

AES helps us understand **why and how perovskite solar cells degrade** by studying **surface element changes** and **chemical reactions**.

It shows **loss of halides, formation of new compounds, and element movement**, helping scientists design **more stable and longer-lasting solar cells**.

6) What are the main challenges of using AES for insulating materials?

Challenges of Using Auger Electron Spectroscopy (AES) for Insulating Materials — very simple explanation

📌 Basic idea

AES works best with **conducting materials**, because it uses an **electron beam** to get information from the surface.

When the material is an **insulator** (like glass, plastic, or oxide), it cannot carry away the extra electrons.

This causes several problems during measurement.

📌 Main challenges (step by step)

1. **Surface charging:**

- When the electron beam hits an insulator, **electrons build up** on the surface because they cannot escape.
- This extra charge changes the **electric field** on the surface.

2. **Peak shifting and distortion:**

- The built-up charge causes the **Auger electron energies** to **shift or blur** in the spectrum.
- It becomes hard to identify which element the peaks belong to.

3. **Beam damage:**

- Continuous electron bombardment can **damage** or **heat up** insulating samples, changing their surface chemistry.

4. **Signal loss:**

- Because of charging, some emitted Auger electrons **cannot reach the detector**, leading to **weak or noisy signals**.
-

❑ Possible solutions (to reduce problems)

- Use a **low-energy or pulsed electron beam** to minimize charging.
 - Use a **thin conductive coating** (like carbon or gold) to let charge escape.
 - Apply a **charge neutralizer** (ion or electron flood gun) to balance surface charge.
 - Work in **low beam current** and **short exposure time** to reduce damage.
-

📌 In short:

The main challenge of using AES on insulators is **surface charging**, which causes **peak shifts**, **signal loss**, and **sample damage**.

Careful control of beam energy, coating, or charge neutralization is needed to get **accurate and stable AES results** on insulating materials.

7) Define photoluminescence (PL), and describe its fundamental mechanism.

Photoluminescence (PL) — very simple explanation

📌 Definition (what is PL)

Photoluminescence (PL) is a process where a material **absorbs light** and then **gives off light** of its own.

It is a simple and **non-destructive** method used to study the **electronic and optical properties** of materials, especially **semiconductors**.

❑ Fundamental mechanism (step by step)

1. **Light absorption:**
 - A **light source** (usually a laser) shines on the material.
 - The light gives energy to **electrons**, making them jump from the **valence band** to the **conduction band**.
 - This creates **electron-hole pairs** (an excited electron and the empty space it left behind).
2. **Relaxation:**
 - The excited electrons may lose a little energy inside the conduction band as heat.
 - After that, they move back toward lower energy levels.
3. **Recombination:**
 - The electrons finally **fall back** to the valence band and **recombine** with holes.
 - When this happens, they **release energy as light** (a photon).
4. **Emission:**
 - The **color or energy** of this emitted light depends on the **energy gap** between the conduction and valence bands.

- This light is collected and analyzed as the **PL spectrum**.
-

📌 What PL tells us

- The **bandgap energy** of the material (from the light color or wavelength).
 - The **quality of the material** (defects or impurities cause extra peaks).
 - The **recombination process** of charge carriers.
-

📌 In short:

Photoluminescence means the material **glows when light shines on it**.

It happens because electrons get excited by light, then fall back and **emit photons**.

By studying this emitted light, we can learn about the material's **bandgap, purity, and defects**.

8) How does photoluminescence spectroscopy help in determining the bandgap of semiconductors?

Photoluminescence (PL) Spectroscopy and Bandgap of Semiconductors — very simple explanation

📌 Basic idea

Photoluminescence spectroscopy helps to **find the bandgap energy** of a semiconductor by studying the **light it emits** after being excited by another light source.

The color or energy of the emitted light tells us the **energy difference between the valence band and conduction band** — this difference is the **bandgap**.

□ How PL spectroscopy works (step by step)

1. A **laser or light source** shines on the semiconductor.
 2. The incoming light gives energy to **electrons**, moving them from the **valence band** to the **conduction band**.
 3. After a short time, the electrons **fall back** to the valence band and **emit light (photon)**.
 4. The **energy** of this emitted light equals the **bandgap energy** (or slightly less if there are defects).
 5. A **detector measures** the wavelength or energy of this emitted light and makes a **PL spectrum** (intensity vs wavelength or energy).
 6. The **main PL peak** in the spectrum corresponds to the **bandgap** of the semiconductor.
-

🔍 How to calculate the bandgap

📊 How to calculate the bandgap

Use the PL peak wavelength (λ) and this formula:

$$E_g(\text{eV}) = \frac{1240}{\lambda \text{ (in nm)}}$$

For example, if the PL peak is at 620 nm, then

$$E_g = \frac{1240}{620} = 2.0 \text{ eV}$$

🔍 What more PL tells us

- Small peaks below the main one show **defects or impurities**.
- The intensity of peaks tells about **material quality** — stronger peaks mean fewer defects.

🔍 In short:

PL spectroscopy helps find the **bandgap** by measuring the **energy of the light** emitted when electrons and holes recombine.

The **main PL peak position** directly shows the **bandgap energy** of the semiconductor.

9) Discuss the applications of photoluminescence in optoelectronic device characterization.

Applications of Photoluminescence (PL) in Optoelectronic Device Characterization — very simple explanation

🔍 Basic idea

Photoluminescence (PL) is used to study the **optical and electronic quality** of materials used in **optoelectronic devices** such as **LEDs, lasers, and solar cells**.

By analyzing the light a material emits, we can learn about its **bandgap, defects, and efficiency** — all very important for device performance.

□ Main applications of PL (step by step)

1. **Bandgap measurement:**
 - PL helps find the **bandgap energy** of materials by measuring the wavelength of emitted light.
 - Knowing the bandgap is essential for designing LEDs, lasers, and solar cells.
2. **Defect detection:**
 - Extra PL peaks at lower energies show the presence of **defects or impurities**.
 - Fewer defects mean **better device quality** and **higher efficiency**.
3. **Material quality check:**
 - Strong PL intensity means the material has **good crystal quality** and **less recombination loss**.
 - Weak PL means more **defects or poor interfaces**.
4. **Carrier lifetime measurement:**
 - Time-resolved PL measures **how long electrons and holes stay excited** before recombining.
 - Longer lifetime means **better charge transport** and **higher efficiency** in devices.
5. **Uniformity mapping:**
 - PL mapping scans the surface to check if the material or thin film is **uniform**.
 - Useful for large wafers used in LEDs or solar cells.
6. **Interface study:**
 - PL helps study **interfaces** between different layers (like perovskite and transport layers) to see if they cause **energy loss**.

📌 In short:

Photoluminescence is a powerful and simple tool to test **bandgap, defects, carrier lifetime, and uniformity** in optoelectronic materials.

It helps improve the performance and reliability of **LEDs, lasers, and solar cells** by showing how good or bad the material quality is.

10) What is Fourier Transform Infrared (FTIR) spectroscopy, and how does it work?

Fourier Transform Infrared (FTIR) Spectroscopy — very simple explanation

📌 Definition (what is FTIR)

FTIR spectroscopy is a **technique used to study chemical bonds and molecular structure** of materials.

It works by passing **infrared (IR) light** through a sample and measuring how much of that light is **absorbed** at different wavelengths.

Each chemical bond absorbs light at specific wavelengths, so the result acts like a **fingerprint** of the material.

□ Working principle (step by step)

1. **Infrared light source:**
 - The FTIR instrument shines a broad range of **infrared light** onto the sample.
 2. **Interaction with the sample:**
 - When IR light hits the sample, **certain wavelengths are absorbed** by the molecules.
 - The absorbed light makes the **bonds in the molecules vibrate** (stretch or bend).
 3. **Interferometer:**
 - Instead of measuring one wavelength at a time, FTIR uses a special device called a **Michelson interferometer**.
 - It mixes all wavelengths of light and produces a signal called an **interferogram**.
 4. **Detection and computer processing:**
 - The detector records this interferogram.
 - A **Fourier Transform (a mathematical process)** separates the mixed signal into individual wavelengths.
 - The final result is an **FTIR spectrum** showing how much light was absorbed at each wavelength.
 5. **Interpretation:**
 - Each **peak** in the spectrum corresponds to a specific **chemical bond or group** (like O–H, C=O, C–H, N–H, etc.).
-

▣ What FTIR tells us

- The **types of chemical bonds** present in a sample.
 - The **functional groups** (like alcohol, carbonyl, or amine).
 - **Chemical composition** and **purity** of materials.
 - Changes in bonding during **reactions, oxidation, or degradation**.
-

▣ In short:

FTIR works by shining **infrared light** on a sample and measuring **which wavelengths are absorbed**.

From the resulting spectrum, we can identify **chemical bonds, molecular structure, and functional groups** — making FTIR a key tool in **chemistry and material analysis**.